

WHITE PAPER

Beyond Ping: Separating Application from Network Issues

How Prisma SD-WAN and Stigix identify the real bottleneck in minutes

Jean-Louis SUZANNE

Technical Sales Manager, Palo Alto Networks

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Tests run with Stigix: <https://github.com/jsuzanne/stigix>

1. The problem everyone recognises

An end user calls the help desk. Their application is slow. The help desk pings the data center: 6 ms, green. They check the SD-WAN dashboard: tunnels up, no alerts. They escalate to the network team, who find nothing. They escalate to the application team, who say their server metrics look fine. Meanwhile, the user is still waiting.

This scenario plays out every day. The reason is simple: a ping, a traceroute or a tunnel health check only measures whether packets can travel between two points. **None of them measures how long the application behind those points takes to answer.** These are different questions, and they need different tools.

The gap. Traditional network monitoring confirms that packets arrived. What it cannot say is whether the server processed those packets in 10 ms or 800 ms. Without that, every slow-application incident starts with the wrong question: "Is it the network?" instead of "What is slow, and where?"

2. What SRT is, and why it matters

Server Response Time (SRT) is the time a server spends processing a request before sending its reply. It is distinct from network latency (RTT), which measures only the transport time between two points.

Metric	What it measures	What it reveals
RTT (Round Trip Time)	Packet travel time: client → server → client	Network transport health
SRT (Server Response Time)	Server processing time: request received → reply sent	Application / server health
App RTT (total)	RTT + SRT	End-to-end user experience

A high App RTT can come from either a slow network (high RTT) or a slow server (high SRT). Without decomposing the two, it is impossible to know which team to call.

Scope: Stigix provides SRT decomposition for **HTTP/REST applications** natively, separating serverDelayMs from networkRttMs at the session level. For all application types including non-HTTP TCP, **Prisma SD-WAN SCM** provides SRT visibility through its own overlay telemetry, as shown in the screenshots later in this paper.

3. Two measurement planes, one platform

Prisma SD-WAN, combined with Stigix, measures both RTT and SRT simultaneously on the same session, without any agent on the server and without decrypting HTTPS traffic.

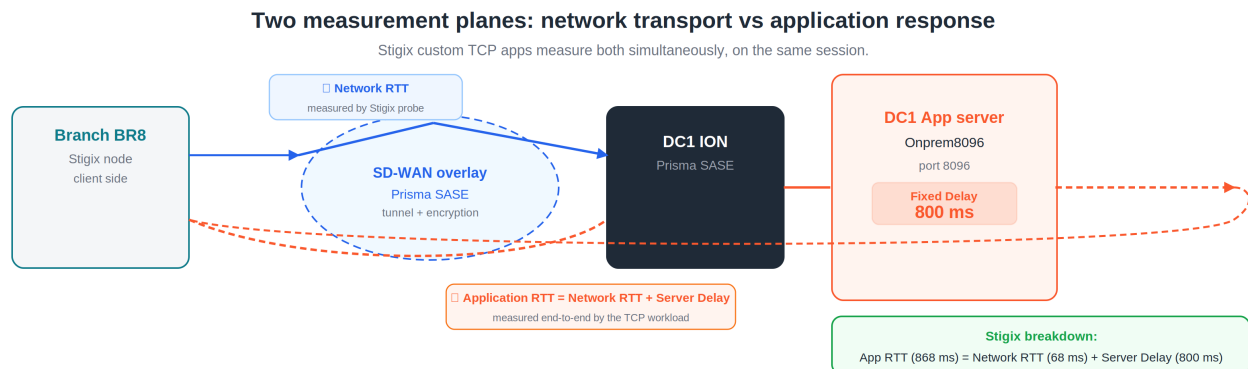


Figure 1. Two measurement planes on the same session. Stigix measures the SD-WAN transport RTT via a convergence probe, and computes SRT from the TCP session timing of the HTTP workload.

The Prisma SD-WAN overlay adds its own measurement layer. The ION devices time every application transaction from the LAN side and report RTT and SRT separately in SCM. Both sources are consistent and complementary: Stigix provides per-session granularity; SCM provides persistent historical trending for all recognized applications.

The power of this approach is that it requires no changes to the application, no agents on the server, and no packet decryption. The decomposition comes from the TCP timing observable at the branch ION.

4. The test: one slow app, one healthy app

- **Onprem8083** (port 8083, HTTP/1.1 REST): echoes every request immediately. SRT $\approx$ 0. This is the healthy baseline.
- **Onprem8096** (port 8096, HTTP/1.1 REST): waits 800 ms before every reply. This simulates a slow backend — a database query that takes too long, an overloaded API, or a misconfigured timeout. The network is not modified.

Server Role / Inbound (Incoming Request Handling)

These settings define how **this node** replies when receiving requests on its local listener port (:8096). *If this node only acts as a Client sending traffic to a remote Data Center, this behavior remains dormant locally (the behavior configured on the remote target server will execute).*

Server Response Simulation Mode

Echo

Replies back with exact payload

Acknowledge

Compact ACK without payload

Fixed Delay

Injects fixed delay before reply

Random Delay

Jittered delay between min/max

Looping Delay

Alternates normal/slow phases

Drop Response

Simulates response loss

Close Connection

Closes socket after N requests

Error Response

Simulates server application errors

EICAR Security Test

Emulates malware payload for AV/NGFW

Simulation Parameters (fixed_delay)

Fixed Delay (ms)

800

Fixed Delay Simulation

Injects a constant synthetic delay on every server response frame. *Simulates predictable high-latency WAN, transcontinental links, or slow backend database processing.*

Back

Next Step

Figure 2. Stigix server behavior: Fixed Delay 800 ms. Every HTTP response is delayed by exactly 800 ms on the server side. The network path is unchanged.

The screenshot shows the 'Custom Applications' interface in the Stigix simulator. At the top, it says 'INTER-SITE & CLOUD SIMULATOR' and describes the simulation capabilities. Below this, there's a 'Local Site: BR8-Ubuntu' and buttons for 'Start All', 'Stop All', 'LIVE', and a refresh icon. A list of applications is shown, each with a status indicator (green for healthy, red for unhealthy) and a 'TX on' count. The applications are: Onprem8083 (Port 8083, HTTP/1.1 REST, TX on (6)), Onprem8084 (Port 8084, TX on (2)), Onprem8085 (Port 8085, TX on (2)), Onprem8086 (Port 8086, TX on (2)), Onprem8087 (Port 8087, TX on (2)), Onprem8088 (Port 8088, TX on (2)), Onprem8089 (Port 8089, TX on (2)), Onprem8090 (Port 8090, TX on (2)), Onprem8091 (Port 8443, TX on (6)), and Onprem8096 (Port 8096, HTTP/1.1 REST, TX on (2)). The Onprem8096 application is highlighted in blue, indicating it is the selected application. Below the list, there are buttons for 'Stop Listener', 'Stop Client', 'Edit Profile', and 'Push to Prisma SD-WAN'.

Figure 3. Stigix custom apps on BR8: Onprem8096 shows 855 ms average App RTT while all other apps are healthy. The anomaly is isolated to one application.

5. What both platforms measured

Stigix: session-level SRT breakdown

Metric	Healthy (8083)	Slow (8096)	What it means
Network RTT (probe)	5.8 ms	5.8 ms	Network unchanged
App RTT p50	10 ms	819 ms	80× slower
serverDelayMs (SRT)	0 ms	800 ms	Server is the bottleneck
networkRttMs	5.8 ms	68 ms	Network is not the issue
Voice MOS	4.4	4.4	Network unaffected
Packet loss	0 %	0 %	Network is healthy

**DC1-Ubuntu**

OUTGOING SESSION



local → 192.168.203.100:8096 • Onprem8096

● **CONNECTED** | ⌚ Uptime: **4h 21m**Jitter: ± **75.92 ms**⚡ **Latency Breakdown (RTT)**

🔄 Flow & Bitrates

📄 Socket & Protocol

⚡ **LATENCY ORIGIN BREAKDOWN****Total App RTT: 810 ms**● **SD-WAN Transport: 10 ms (1%)**● **Server Processing Delay: 800 ms (99%)**

⚠ **Simulated Server Delay Active:** The remote server behavior adds a fixed or random artificial processing delay of **800ms**. The true network path transit is **10ms**.

MINIMUM

806 ms

MEDIAN (P50)

817 ms

TAIL (P95)

1094 ms

PEAK (MAX)

1570 ms📊 **Real-Time RTT Waveform (Last 20 ticks)**

Avg: 860.15 ms

📄 **Understanding Latency Metrics & Percentiles**

Median (P50): 50% of requests are faster than this value. Represents typical baseline performance.

Minimum: Best observed physical transport floor across underlay/overlay tunnels.

Tail Latency (P95): 95% of requests complete within this delay. Highlights SD-WAN tunnel jitter and micro-burst queuing.

Peak (Max): Worst-case outlier transaction recorded in the sliding window.

Real-time telemetry stream active (1.5s refresh)

⚡ **Send SYN/Handshake Probe****Close Drawer**

Figure 4. Stigix Latency Origin Breakdown: SD-WAN Transport 10 ms (1%) vs Server Processing Delay 800 ms (99%). Total App RTT: 810 ms.

Prisma SD-WAN SCM: the same picture from the overlay

Prisma SD-WAN SCM surfaces the same diagnosis independently, using its own overlay telemetry. The application STX_Onprem8096 is visible in SCM with full SRT and RTT trending:

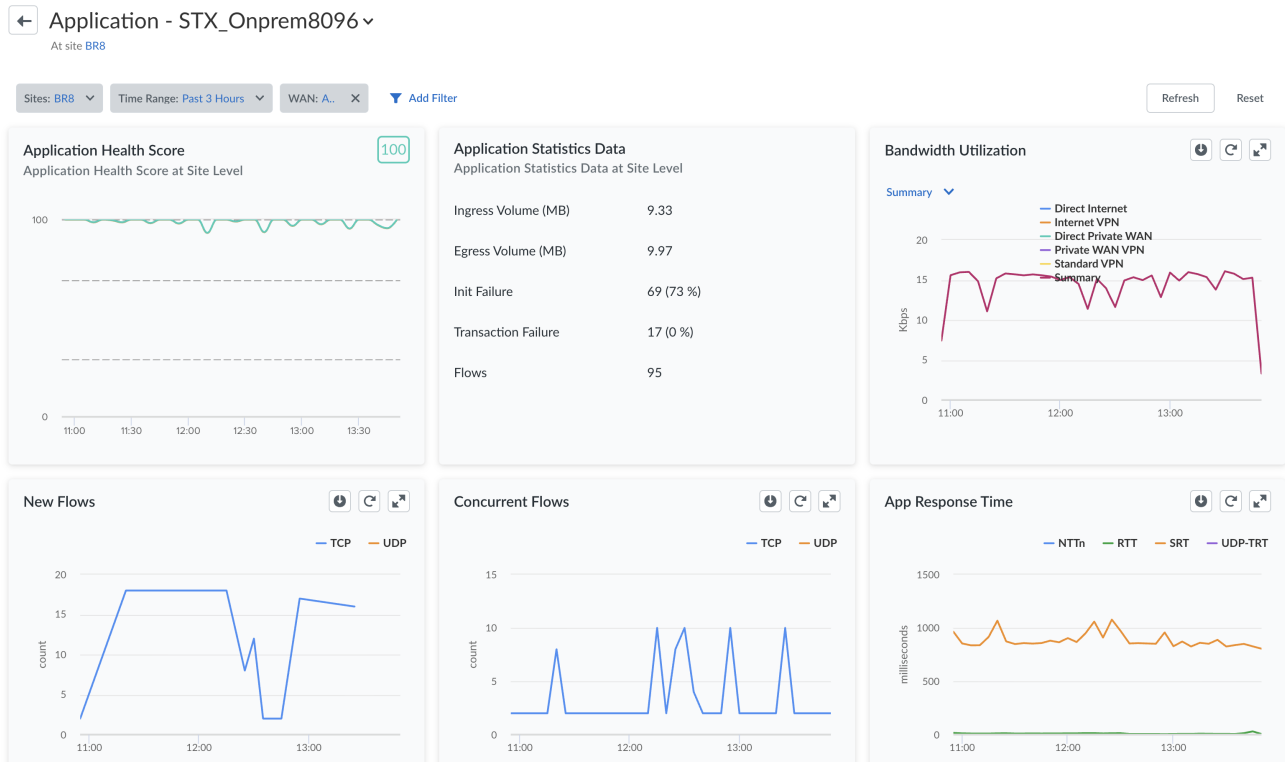


Figure 5. SCM Application dashboard — STX_Onprem8096 at site BR8. Health Score: 100/100. App Response Time (SRT, orange) consistently at 800–1 000 ms. RTT (green) flat near 0. The SD-WAN path is healthy; the server delay is the anomaly.

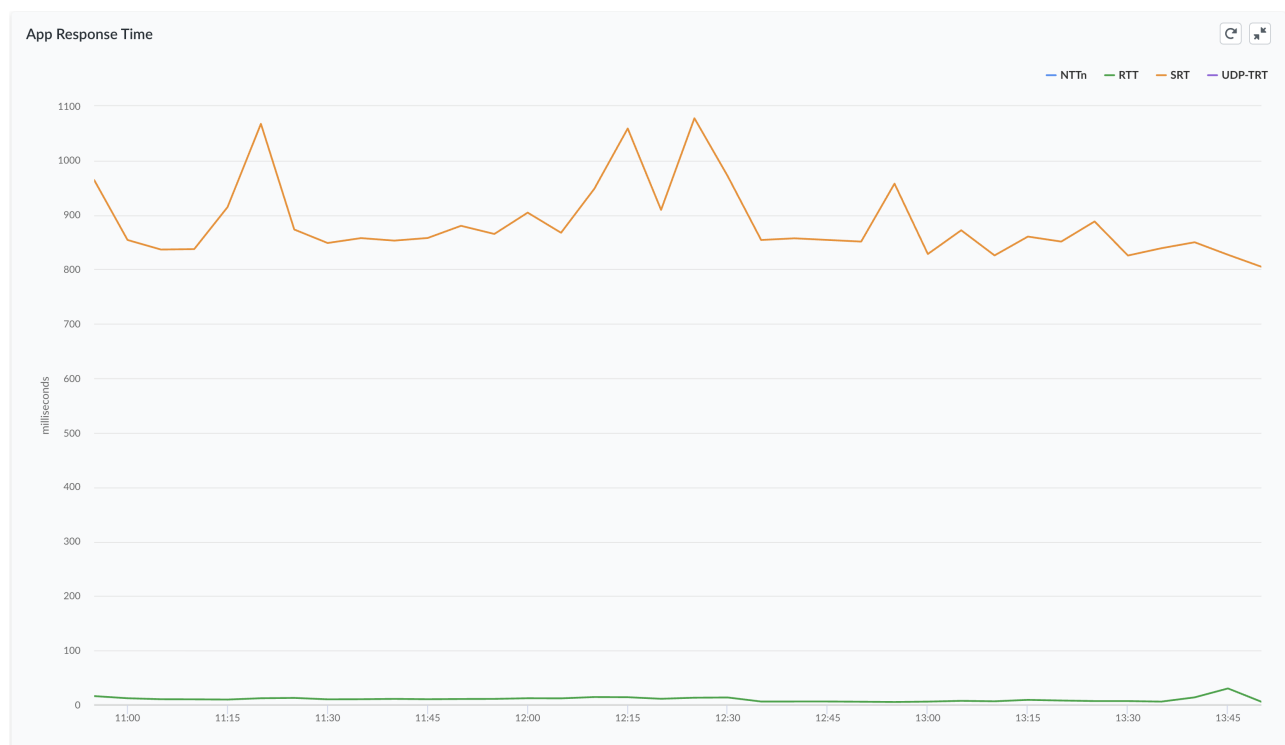


Figure 6. SCM App Response Time detail over 3 hours: SRT (orange) at 800–1 100 ms. RTT (green) below 20 ms throughout. This trending confirms the problem is persistent, not a transient spike.

Prisma SD-WAN Flow Browser: path confirmed

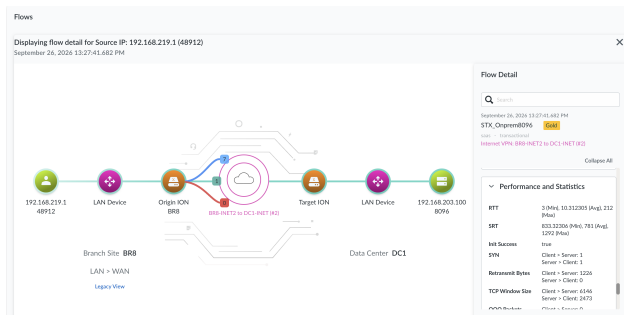


Figure 7. Flow Browser: STX_Onprem8096 on BR8-INET2 → DC1-INET. Path stable, no changes.

Performance and Statistics

RTT	3 (Min), 10.312305 (Avg), 212 (Max)
SRT	833.32306 (Min), 781 (Avg), 1292 (Max)
Init Success	true
SYN	Client > Server: 1 Server > Client: 1
Retransmit Bytes	Client > Server: 1226 Server > Client: 0
TCP Window Size	Client > Server: 6146 Server > Client: 2473

Figure 8. SCM flow stats: RTT avg 10 ms, SRT avg 781 ms. The gap is the server delay.

Two independent tools, same conclusion. Stigix: serverDelayMs = 800 ms, networkRttMs = 68 ms. Prisma SCM: SRT = 781 ms avg, RTT = 10 ms avg. Both point to the same root cause: the server, not the network.

The diagnostic in one picture: network RTT vs application RTT

Same destination (DC1), three different states. The breakdown tells the story.

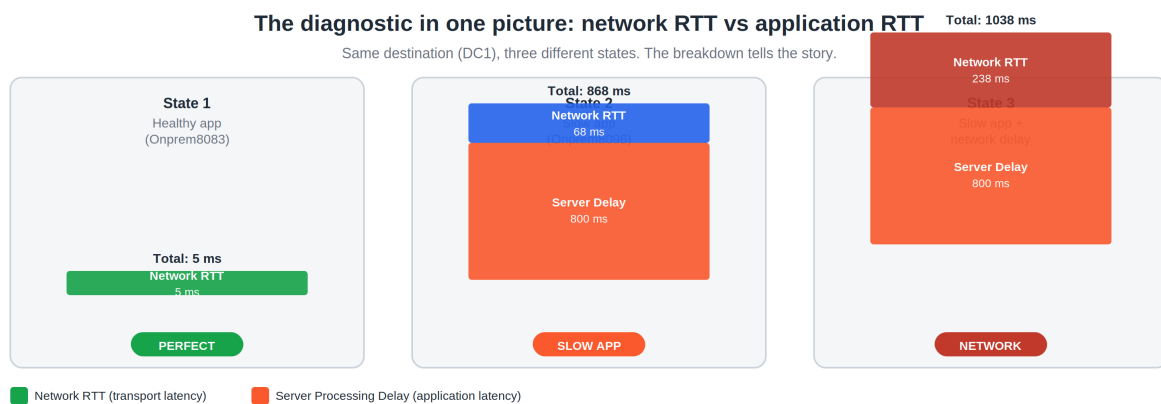


Figure 9. The diagnostic in one chart: three states, same destination. The breakdown of each bar tells the story at a glance.

6. Counter-example: what a network problem looks like

+200 ms of latency was injected on the underlay circuit (eth10, DC1-INET-221). The contrast with the slow-application case is immediate and unmistakable:

Signal	Slow app (Onprem8096)	Network degradation (+200 ms)
Convergence probe RTT	5.8 ms — PERFECT	93 ms — degraded
Apps affected	Onprem8096 only	All apps toward DC1
SRT / serverDelayMs	800 ms	800 ms — unchanged
networkRttMs	68 ms	238 ms — +250%
Voice MOS	4.4 — unaffected	Degraded
Packet loss	0 %	0.4%, blackout 183 ms
Diagnosis	Server / application	Network / transport

The two signatures never overlap. A slow application: one app affected, probe unchanged, SRT dominates. A network degradation: all apps affected simultaneously, probe climbs, SRT stays constant while RTT rises.

7. Diagnostic logic: two checks, two minutes

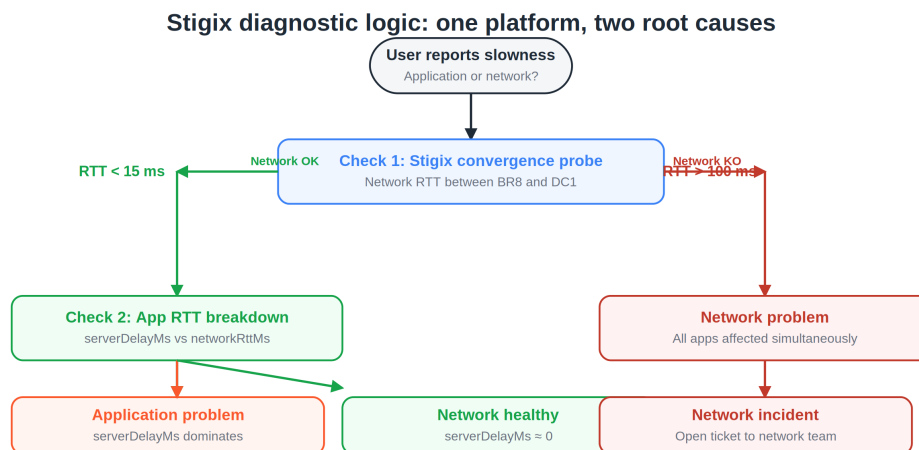


Figure 10. The diagnostic flow. Two checks, two minutes, unambiguous verdict.

- 1 Check 1 — Convergence probe:** network RTT normal? If high or lossy, it is a network incident. Stop here.
- 2 Check 2 — SRT breakdown:** for HTTP apps, use Stigix serverDelayMs. For all apps, use Prisma SCM SRT dashboard. If SRT dominates, escalate to the app team.

8. Recommendations

- 1 Always decompose RTT from SRT.** Use Stigix for HTTP app sessions; use Prisma SCM for all application types. Never rely on ping alone.
- 2 Use both tools together.** Stigix provides real-time session-level detail. SCM provides persistent trending and covers non-HTTP applications natively.
- 3 Run the counter-example in demos.** Showing what a network problem looks like — all apps affected, SRT unchanged — is what makes the method credible.
- 4 Baseline your SRT.** Record healthy SRT for each critical application. Alert on 3× deviation.
- 5 Voice MOS is a free health indicator.** If MOS stays at 4.4 while one application is slow, the network is almost certainly innocent.

Measurement notes

SRT decomposition in Stigix (serverDelayMs / networkRttMs) requires the application to be configured as HTTP/REST. For plain TCP applications, Stigix reports total App RTT; use Prisma SCM for SRT visibility on non-HTTP traffic.

The 68 ms average networkRttMs differs from the 5.8 ms convergence probe because it is averaged over the full session lifetime including TCP connection setup.

The +200 ms counter-example was applied on the underlay circuit (eth10), not the SD-WAN overlay. The probe saw 93 ms — not 200 ms — because the SD-WAN had two underlay paths.